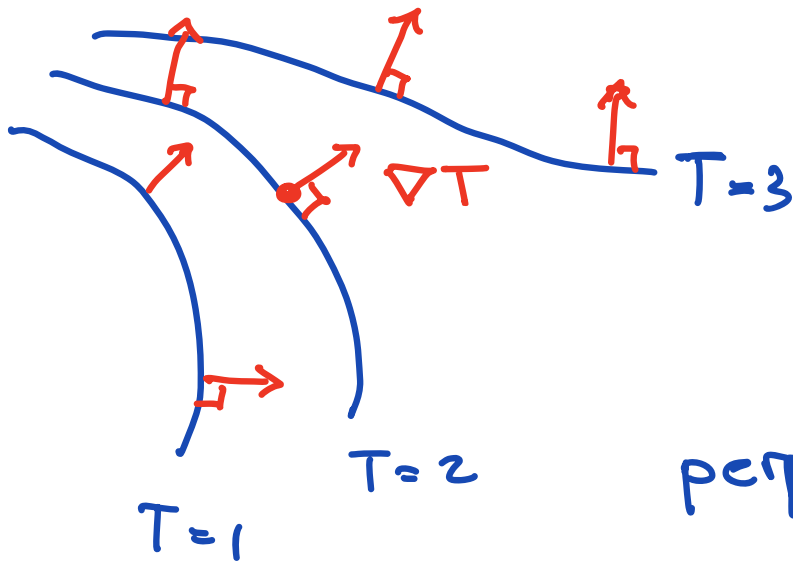


Gradient & Directional derivative

Gradient:



$$\nabla T = \begin{bmatrix} \frac{\partial T}{\partial x} \\ \frac{\partial T}{\partial y} \\ \frac{\partial T}{\partial z} \end{bmatrix}$$

perp. to T contours
points uphill

Gradient: scalar $\rightarrow$ vector

$$\nabla T = \underline{v}$$

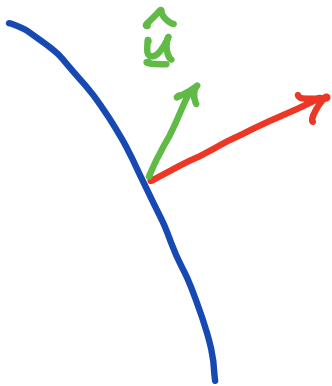
Divergence: vector $\rightarrow$ scalar

$$\nabla \cdot \underline{v} = \alpha$$

Change of T in direction $\hat{u}$?

$\nabla T \Rightarrow$ directional derivative

$$D_{\hat{u}} T = \nabla T \cdot \underline{u} = \text{scalar?}$$



$\Rightarrow$ How do we extend these concepts
to $\nabla \underline{v}$ and $\nabla \cdot \underline{S}$?

Differentiation of Tensor fields

A field is a function of space.

scalar fields: $\phi(\underline{x})$ temp., density

vector fields: $\underline{v}(\underline{x})$ force, velocity

tensor fields: $\underline{\underline{S}}(\underline{x})$ stress, conductivity

Today's lecture is review and extension of concepts from multivariable calculus.

Gradients

Gradient of scalar field

Scalar field $\phi(\underline{x})$ is differentiable at $\underline{x}$

if there exists a vector field $\nabla\phi \in \mathcal{V}$ such that

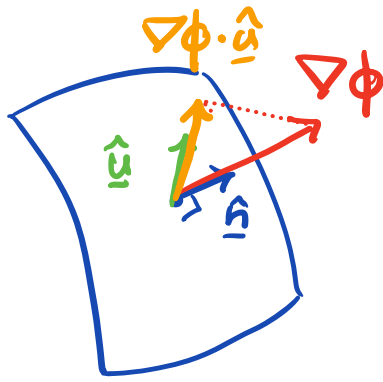
$$\phi(\underline{x} + \underline{h}) = \phi(\underline{x}) + \nabla\phi(\underline{x}) \cdot \underline{h} + \mathcal{O}(|\underline{h}|)$$

by Taylor expansion. Or equivalently

$$\nabla\phi(\underline{x}) \cdot \underline{\hat{u}} = \left. \frac{d}{d\varepsilon} \phi(\underline{x} + \varepsilon \underline{\hat{u}}) \right|_{\varepsilon=0} \quad \text{for all } \underline{v} \in \mathcal{V}$$

where $\underline{h} = \varepsilon \underline{\hat{u}}$ and $|\underline{\hat{u}}| = 1$.

→ The vector $\nabla\phi$ is called the gradient of ϕ .



Consider a level set of ϕ

$\nabla\phi \parallel \underline{\hat{n}}$ in direction of increasing ϕ

$$\underline{\hat{n}} = \frac{\nabla\phi}{|\nabla\phi|}$$

Directional derivative (Gâteaux operator)

$$D_{\underline{\hat{u}}}\phi(\underline{x}) = \left. \frac{d}{d\epsilon} \phi(\underline{x} + \epsilon \underline{\hat{u}}) \right|_{\epsilon=0} = \nabla\phi(\underline{x}) \cdot \underline{\hat{u}}$$

Representation of the gradient in frame $\{\underline{e}_i\}$

$$\phi(\underline{\bar{x}} + \epsilon \underline{u}) = \phi(\underbrace{\bar{x}_1 + \epsilon u_1}_{x_1}, \underbrace{\bar{x}_2 + \epsilon u_2}_{x_2}, \underbrace{\bar{x}_3 + \epsilon u_3}_{x_3})$$

$$\nabla\phi \cdot \underline{\hat{u}} = \left. \frac{d}{d\epsilon} \phi(\bar{x}_1 + \epsilon u_1, \bar{x}_2 + \epsilon u_2, \bar{x}_3 + \epsilon u_3) \right|_{\epsilon=0}$$

$$= \frac{d\phi}{dx_1} \frac{dx_1}{d\epsilon} + \frac{d\phi}{dx_2} \frac{dx_2}{d\epsilon} + \frac{d\phi}{dx_3} \frac{dx_3}{d\epsilon} \Big|_{\epsilon=0}$$

$$= \frac{d\phi}{dx_1} u_1 + \frac{d\phi}{dx_2} u_2 + \frac{d\phi}{dx_3} u_3$$

$$= \frac{\partial\phi}{\partial x_i} u_i = \phi_{,i} u_i = \phi_{,i} u_j \delta_{ij} = \phi_{,i} u_j \underline{e}_i \cdot \underline{e}_j$$

$$\nabla\phi \cdot \underline{\hat{u}} = (\phi_{,i} \underline{e}_i) \cdot (u_j \underline{e}_j) \quad \checkmark$$

Note: Index notation for derivatives

$$\boxed{\frac{\partial \phi}{\partial x_i} = \phi_{,i}} \quad \text{derivative index after comma!}$$

Gradient in components: $[\nabla \phi] = \phi_{,i} \underline{e}_i = \begin{pmatrix} \frac{\partial \phi}{\partial x_1} \\ \frac{\partial \phi}{\partial x_2} \\ \frac{\partial \phi}{\partial x_3} \end{pmatrix}$

Heat conduction:

potential: $T(\underline{x})$ Temperature

heat flux: $\underline{q} = - \underset{\substack{\uparrow \\ \text{thermal conductivity } [\frac{W}{mK}]}]{k} \nabla T$ (Fourier's law)

Molecular diffusion:

potential: $c(\underline{x})$ Concentration

diffusive flux: $\underline{j} = - \underset{\substack{\uparrow \\ \text{molecular diffusivity } [\frac{m^2}{s}]}]{D} \nabla c$ (Fick's law)

Groundwater flow:

potential: $h(\underline{x})$ hydraulic head

volumetric flux: $\underline{q} = - \underset{\substack{\uparrow \\ \text{hydraulic conductivity } [\frac{m}{s}]}]{K} \nabla h$ (Darcy's law)

Gravitational field: $\underline{g} = - \nabla \Phi$

Gradient of a vector field

A vector field $\underline{v}(\underline{x}) \in \mathcal{V}$ is differentiable at $\underline{x}$ if there exists a tensor field $\nabla \underline{v}(\underline{x}) \in \mathcal{V}^2$ such that

$$\underline{v}(\underline{x} + \underline{h}) = \underline{v}(\underline{x}) + \nabla \underline{v}(\underline{x}) \underline{h} + o(|\underline{h}|)$$

by Taylor expansion or equivalently

$$\nabla \underline{v} \hat{\underline{u}} = \left. \frac{d}{d\epsilon} \underline{v}(\underline{x} + \epsilon \hat{\underline{u}}) \right|_{\epsilon=0} \quad \text{for all } \underline{u} \in \mathcal{V}$$

where $\underline{h} = \epsilon \hat{\underline{u}}$

In frame $\{\underline{e}_j\}$ we write components of $\underline{v}$

as $v_i = v_i(x_1, x_2, x_3)$. For any scalar ϵ

and unit vector $\hat{\underline{u}} = u_k \underline{e}_k$ at $\bar{\underline{x}} = \bar{x}_j \underline{e}_j$

we have the i -th component

$$v_i(\bar{\underline{x}} + \epsilon \hat{\underline{u}}) = v_i(\bar{x}_1 + \epsilon u_1, \bar{x}_2 + \epsilon u_2, \bar{x}_3 + \epsilon u_3)$$

by the chain rule

$$\frac{d}{d\epsilon} v_i(\bar{\underline{x}} + \epsilon \hat{\underline{u}}) = \frac{\partial v_i}{\partial x_1} u_1 + \frac{\partial v_i}{\partial x_2} u_2 + \frac{\partial v_i}{\partial x_3} u_3 = \frac{\partial v_i}{\partial x_j} u_j$$

For full vector $\underline{v} = v_i \underline{e}_i$

$$\begin{aligned}\nabla_{\underline{v}} \hat{\underline{u}} &= \frac{d}{d\varepsilon} \underline{v}(\bar{\underline{x}} + \varepsilon \hat{\underline{u}}) \Big|_{\varepsilon=0} = \frac{d}{d\varepsilon} (v_i(\bar{\underline{x}} + \varepsilon \hat{\underline{u}}) \underline{e}_i) \Big|_{\varepsilon=0} \\ &= \frac{d}{d\varepsilon} (v_i(\bar{\underline{x}} + \varepsilon \hat{\underline{u}})) \Big|_{\varepsilon=0} \underline{e}_i = \frac{\partial v_i}{\partial x_j} u_j \underline{e}_i\end{aligned}$$

components: $[\nabla_{\underline{v}}]_{ij} = \frac{\partial v_i}{\partial x_j} = v_{i,j}$

Representation $\nabla_{\underline{v}} = v_{i,j} \underline{e}_i \otimes \underline{e}_j$

Explicit

$$\begin{aligned}\nabla_{\underline{v}} &= \begin{bmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{bmatrix} = \begin{bmatrix} \nabla_{v_1}^T \\ \nabla_{v_2}^T \\ \nabla_{v_3}^T \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial \underline{v}}{\partial x_1} & \frac{\partial \underline{v}}{\partial x_2} & \frac{\partial \underline{v}}{\partial x_3} \end{bmatrix}\end{aligned}$$

Examples:

1) Strain tensor: $\underline{\underline{\varepsilon}} = \text{sym}(\nabla \underline{u}) = \frac{1}{2} (\nabla \underline{u} + \nabla \underline{u}^T)$

2) Rate of strain tensor: $\underline{\underline{\dot{\varepsilon}}} = \frac{1}{2} (\nabla \underline{v} + \nabla \underline{v}^T)$

$\underline{u}$ = displacement $\underline{v}$ = velocity

Divergence of a vector field

Def: To any $\underline{v}(\underline{x}) \in \mathcal{V}$ we associate a scalar field $\nabla \cdot \underline{v}$ called the divergence of $\underline{v}$

$$\nabla \cdot \underline{v} = \text{tr}(\nabla \underline{v})$$

In frame $\{\underline{e}_i\}$ with $\underline{v}(\underline{x}) = v_i(\underline{x}) \underline{e}_i$ we have

$$\nabla \cdot \underline{v} = \text{tr}(\nabla \underline{v}) = v_{i,i} = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3}$$

If $\nabla \cdot \underline{v} = 0$ a field is solenoidal or divergence free. If $\underline{v}$ is a displacement or velocity then $\nabla \cdot \underline{v}$ is related to (rate of) volume change.

Examples:

Gauss' law of gravity: $\nabla \cdot \underline{g} = -4\pi G\rho$

Continuity condition: $\nabla \cdot \underline{v} = 0$

(incompressible flows)

Divergence of a tensor field

To any $\underline{\underline{S}}(\underline{x}) \in \mathcal{V}^2$ we associate a vector field $\nabla \cdot \underline{\underline{S}} \in \mathcal{V}$ called the divergence of $\underline{\underline{S}}$

$$(\nabla \cdot \underline{\underline{S}}) \cdot \underline{a} = \nabla \cdot (\underline{\underline{S}}^T \underline{a}) \quad \text{for all } \underline{a} \in \mathcal{V}$$

uses definition of vector divergence!

In frame $\{\underline{e}_i\}$ with $\underline{\underline{S}} = S_{ij} \underline{e}_i \otimes \underline{e}_j$ and $\underline{a} = a_k \underline{e}_k$ we have $\underline{q} = \underline{\underline{S}}^T \underline{a}$ or $q_j = S_{ij} a_i$ ($q_i = S_{ji} a_j$)

substituting

$$\begin{aligned} (\nabla \cdot \underline{\underline{S}}) \cdot \underline{a} &= \nabla \cdot (\underline{\underline{S}}^T \underline{a}) = \nabla \cdot \underline{q} = \text{tr}(\nabla \underline{q}) = q_{j,i} \\ &= S_{ij,j} a_i = (S_{ij,j} \underline{e}_i) \cdot (a_k \underline{e}_k) \end{aligned}$$

by the arbitrariness of $\underline{a}$ we have

$$\nabla \cdot \underline{\underline{S}} = S_{ij,j} \underline{e}_i$$

Gradient & Divergence product rules

$$\nabla \cdot (\phi \underline{v}) = \underline{v} \cdot \nabla \phi + \phi \nabla \cdot \underline{v}$$

$$\phi \in \mathbb{R}$$

$$\nabla \cdot (\phi \underline{\underline{s}}) = \underline{\underline{s}} \cdot \nabla \phi + \phi \nabla \cdot \underline{\underline{s}}$$

$$\underline{v} \in \mathcal{V},$$

$$\nabla \cdot (\underline{\underline{s}}^T \underline{v}) = (\nabla \cdot \underline{\underline{s}}) \cdot \underline{v} + \underline{\underline{s}} : \nabla \underline{v}$$

$$\underline{\underline{s}} \in \mathcal{V}^2$$

$$\nabla(\phi \underline{v}) = \underline{v} \otimes \nabla \phi + \phi \nabla \underline{v}$$

Example: $\nabla \cdot (\underline{\underline{s}}^T \underline{v})$ note $\underline{\underline{s}} = \underline{\underline{s}}(\underline{x})$ and $\underline{v} = \underline{v}(\underline{x})$

$$q(\underline{x}) = \underline{\underline{s}}^T(\underline{x}) \underline{v}(\underline{x})$$

$$q_j = s_{ij} v_i$$

$$\nabla \cdot q = \text{tr}(q) = q_{j,j} = (s_{ij} v_i)_{,j}$$

$$= s_{ij,j} v_i + s_{ij} v_{i,j}$$

$$= (\nabla \cdot \underline{\underline{s}}) \cdot \underline{v} + \underline{\underline{s}} : \nabla \underline{v} \quad \checkmark$$

→ useful for energy balance!

$$\text{Example: } \nabla(\phi \underline{v}) = (\phi v_i)_{,j} \underline{e}_i \otimes \underline{e}_j$$

$$= (\phi_{,j} v_i + \phi v_{i,j}) \underline{e}_i \otimes \underline{e}_j$$

$$= v_i \phi_{,j} \underline{e}_i \otimes \underline{e}_j + \phi v_{i,j} \underline{e}_i \otimes \underline{e}_j$$

$$= \underline{v} \otimes \nabla \phi + \phi \nabla \underline{v} \quad \checkmark$$

Laplacian

I) Laplacian of scalar field

$$\Delta\phi = \nabla^2\phi = \nabla \cdot \nabla\phi$$

In frame $\{\underline{e}_i\}$ with $\nabla\phi = \phi_{,i}\underline{e}_i$ we have

$$\nabla \cdot \nabla\phi = \text{tr}(\nabla\nabla\phi) = \text{tr}(\phi_{,ij}\underline{e}_i \otimes \underline{e}_j) = \phi_{,ii}$$

$$\nabla^2\phi = \phi_{,ii}$$

Scalar Laplacian governs steady heat flow.

Example: Poisson's equation for gravity

1) Grav. field eqn: $\underline{g} = -\nabla\Phi$

2) Gauss' law: $\nabla \cdot \underline{g} = -4\pi G\rho$

$$-\nabla \cdot \nabla\Phi = -4\pi G\rho$$

$$\nabla^2\Phi = 4\pi G\rho$$

II Laplacian of vector field

$$\Delta \underline{v} = \nabla^2 \underline{v} = \nabla \cdot \nabla \underline{v}$$

in index notation:

$$\underline{v} = v_i \underline{e}_i, \quad \nabla \underline{v} = v_{i,j} \underline{e}_i \otimes \underline{e}_j \text{ and } \nabla \cdot \underline{s} = s_{ij,j} \underline{e}_i$$

$$\Delta \underline{v} = v_{i,jj} \underline{e}_i$$

Example: Stokes flow (creeping flow)

$$\mu \nabla^2 \underline{v} = \nabla p$$

μ = viscosity

$$\nabla \cdot \underline{u} = 0$$

p = pressure